

Binomial expansion - Set 2 - Solutions

Specification MM2.4.

Answer key

Question	Answer	Topic
1	H	Binomial expansion
2	C	Binomial expansion
3	H	Binomial expansion
4	B	Binomial expansion

Worked solutions

Question 1 BIN-12 · Binomial expansion · spec MM2.4

The expansion of $\left(2x - \frac{1}{x^2}\right)^9$ contains a term in x^k for each of which of the following values of k ?

I $k = 0$

II $k = 3$

III $k = -6$

Answer: H

General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$ has power $9 - r - 2r = 9 - 3r$, which takes values 9, 6, 3, 0, $-3, \dots, -18$. $k = 0$ needs $r = 3$: yes. $k = 3$ needs $r = 2$: yes. $k = -6$ needs $r = 5$: yes. All three.

Full method - how and why

Which powers of x actually occur. Find the general power as an expression in r and see which target values it can hit.

1 General term: $\binom{9}{r}(2x)^{9-r}(-x^{-2})^r$, whose power of x is $(9 - r) + (-2r) = 9 - 3r$.

why: Add the powers contributed by each bracket; the $-\frac{1}{x^2}$ contributes -2 per copy.

2 As r runs from 0 to 9, the power takes the values 9, 6, 3, 0, $-3, -6, -9, -12, -15, -18$.

why: They step down by 3 - so only multiples of 3 appear, all the way down to -18 .

3 $k = 0$ needs $r = 3$ ✓; $k = 3$ needs $r = 2$ ✓; $k = -6$ needs $r = 5$ ✓.

why: Each value of r must be a whole number between 0 and 9 - that is the only restriction.

4 All three occur: 'I, II and III'.

why: Solve $9 - 3r = k$ for each target and check that r is a valid index.

Where it goes wrong

- Forgetting that the second bracket contributes x^{-2r} , not x^{-r} .
- Assuming negative powers cannot appear because it is a 'binomial expansion'.

Takeaway Express the power of x as a linear function of r ; a term exists exactly when the required r is an integer in range.

Question 2 BIN-15 · Binomial expansion · spec MM2.4

The coefficient of x^2 in $(1+x)^n + (1+x)^{n+1}$ is 36. Find n .

Answer: C

$$\binom{n}{2} + \binom{n+1}{2} = \frac{n(n-1) + (n+1)n}{2} = n^2 = 36, \text{ so } n = 6.$$

Full method - how and why

Two expansions added. Write both coefficients in factorial form and the sum simplifies remarkably.

- 1 The x^2 coefficients are $\binom{n}{2}$ and $\binom{n+1}{2}$.

why: One from each expansion; the total coefficient of x^2 is their sum.

- 2 Write them out: $\frac{n(n-1)}{2} + \frac{(n+1)n}{2} = \frac{n[(n-1) + (n+1)]}{2}$.

why: Factor out the $\frac{n}{2}$ - the brackets then collapse.

- 3 Simplify: $\frac{n \times 2n}{2} = n^2$.

why: A surprisingly clean result; recognising it makes the final step trivial.

- 4 Solve $n^2 = 36$, so $n = 6$ (positive integer).

why: n counts an index, so the negative root is rejected.

Where it goes wrong

- Using Pascal's rule to write the sum as $\binom{n+1}{3}$ - that rule adds **adjacent** entries in the same row, not the same entry in different rows.
- Taking $n = \pm 6$ and offering -6 .

Takeaway $\binom{n}{2} + \binom{n+1}{2} = n^2$: write coefficients in factorial form and look for common factors before solving.

Question 3 X2-13 · Binomial expansion · spec MM2.4

n is a positive integer. Which of the following statements is/are true?

I The coefficient of x^n in $(1+x)^{2n}$ is $\binom{2n}{n}$.

II The sum of the coefficients in the expansion of $(1-x)^n$ is 0.

III $(1+x)^n \geq 1+nx$ for all $x \geq 0$.

Answer: H

I is the general term with $r = n$. II: put $x = 1$. III: expanding $(1+x)^n$ gives $1+nx$ plus non-negative terms when $x \geq 0$.

Full method - how and why

Three binomial facts: the general term, substituting a value, and a term-by-term inequality.

- 1 I: the coefficient of x^r in $(1+x)^m$ is $\binom{m}{r}$. With $m = 2n$ and $r = n$ that is $\binom{2n}{n}$. **True.**

why: Straight from the binomial theorem.

- 2 II: the sum of the coefficients of any polynomial is its value at $x = 1$. Here $(1-1)^n = 0^n = 0$ for $n \geq 1$. **True.**

why: Substituting $x = 1$ makes every power equal 1, so the value is the coefficient sum.

- 3 III: for $x \geq 0$, $(1+x)^n = 1+nx + \binom{n}{2}x^2 + \dots$, and every omitted term is non-negative. So $(1+x)^n \geq 1+nx$.

True.

why: All binomial coefficients and all powers of a non-negative x are non-negative.

- 4 So all three.

why: Match to the list.

Where it goes wrong

- Thinking II fails for $n = 0$ - the question says n is a positive integer.
- Doubting III for large n ; the extra terms only help, since $x \geq 0$.
- Confusing $\binom{2n}{n}$ with $\binom{2n}{2}$ or $2\binom{n}{2}$.

Takeaway Sum of coefficients = value at $x = 1$; and $(1+x)^n \geq 1+nx$ for $x \geq 0$ (Bernoulli) follows from discarding non-negative terms.

Question 4 BIN-14 · Binomial expansion · spec MM2.4

How many terms in the expansion of $(x + y + z)^4$ (after collecting like terms) are there?

Answer: B

Terms are $x^a y^b z^c$ with $a + b + c = 4$, $a, b, c \geq 0$. Counting: for $a = 0$ there are 5 choices of (b, c) , $a = 1$: 4, ..., $a = 4$: 1. Total $5 + 4 + 3 + 2 + 1 = 15$.

Full method - how and why

A trinomial expansion. Counting the distinct terms is a counting problem: how many ways can the powers add to 4?

1 Every term has the form $x^a y^b z^c$ with $a + b + c = 4$ and $a, b, c \geq 0$.

why: Each of the four brackets contributes exactly one of x , y or z , so the powers add to 4.

2 Count systematically by the value of a : if $a = 0$ then $b + c = 4$, giving 5 pairs; $a = 1$ gives 4; $a = 2$ gives 3; $a = 3$ gives 2; $a = 4$ gives 1.

why: For a fixed a , the number of (b, c) pairs is $(4 - a) + 1$.

3 Total: $5 + 4 + 3 + 2 + 1 = 15$.

why: A triangular number - the standard result $\binom{n+2}{2}$ for three variables, here $\binom{6}{2} = 15$.

Where it goes wrong

- Answering $3^4 = 81$ - that counts the terms **before** collecting like terms.
- Answering 5 by counting only the powers of x .

Takeaway Distinct terms of $(x + y + z)^n$ correspond to solutions of $a + b + c = n$ in non-negative integers: $\binom{n+2}{2}$ of them.